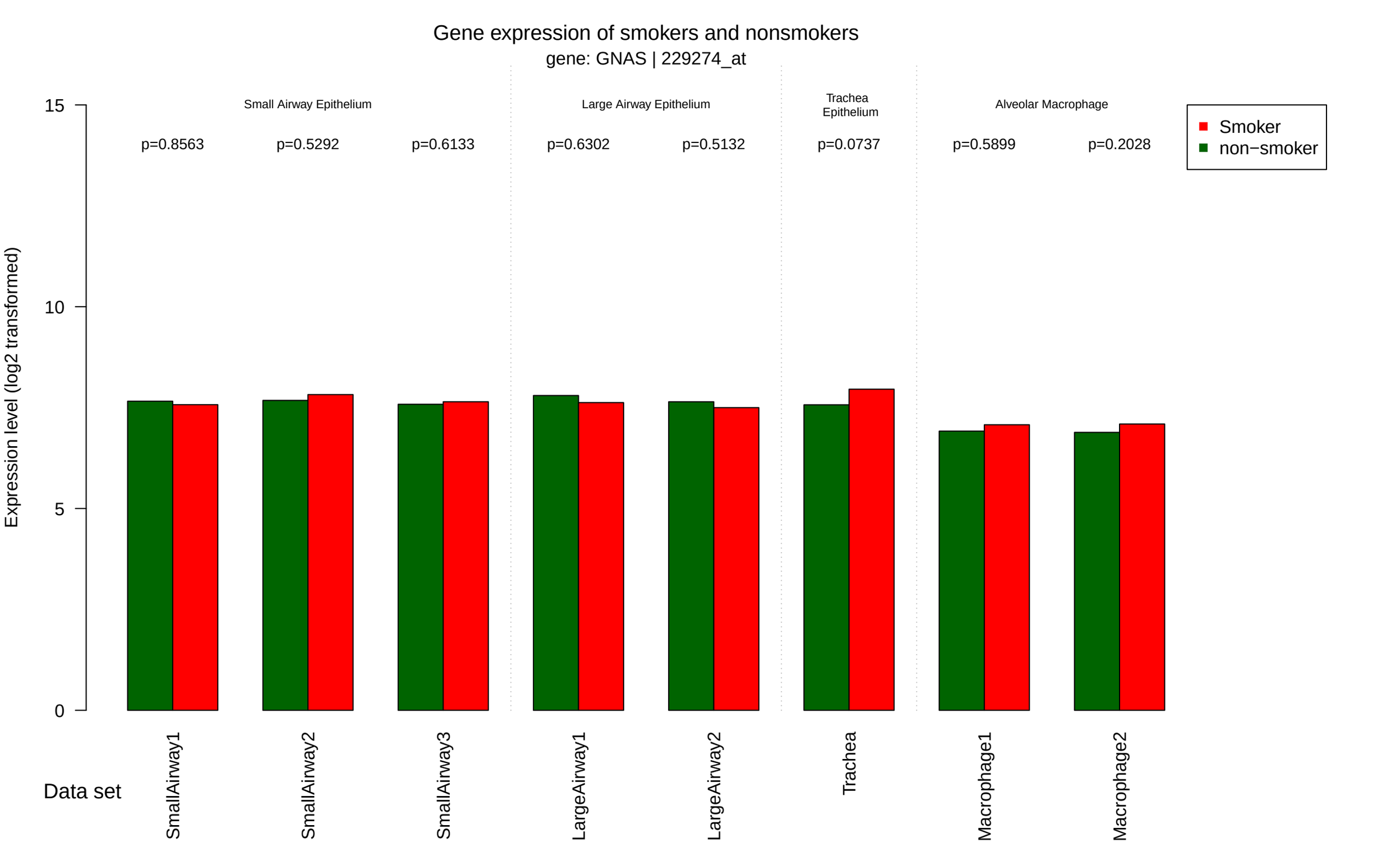
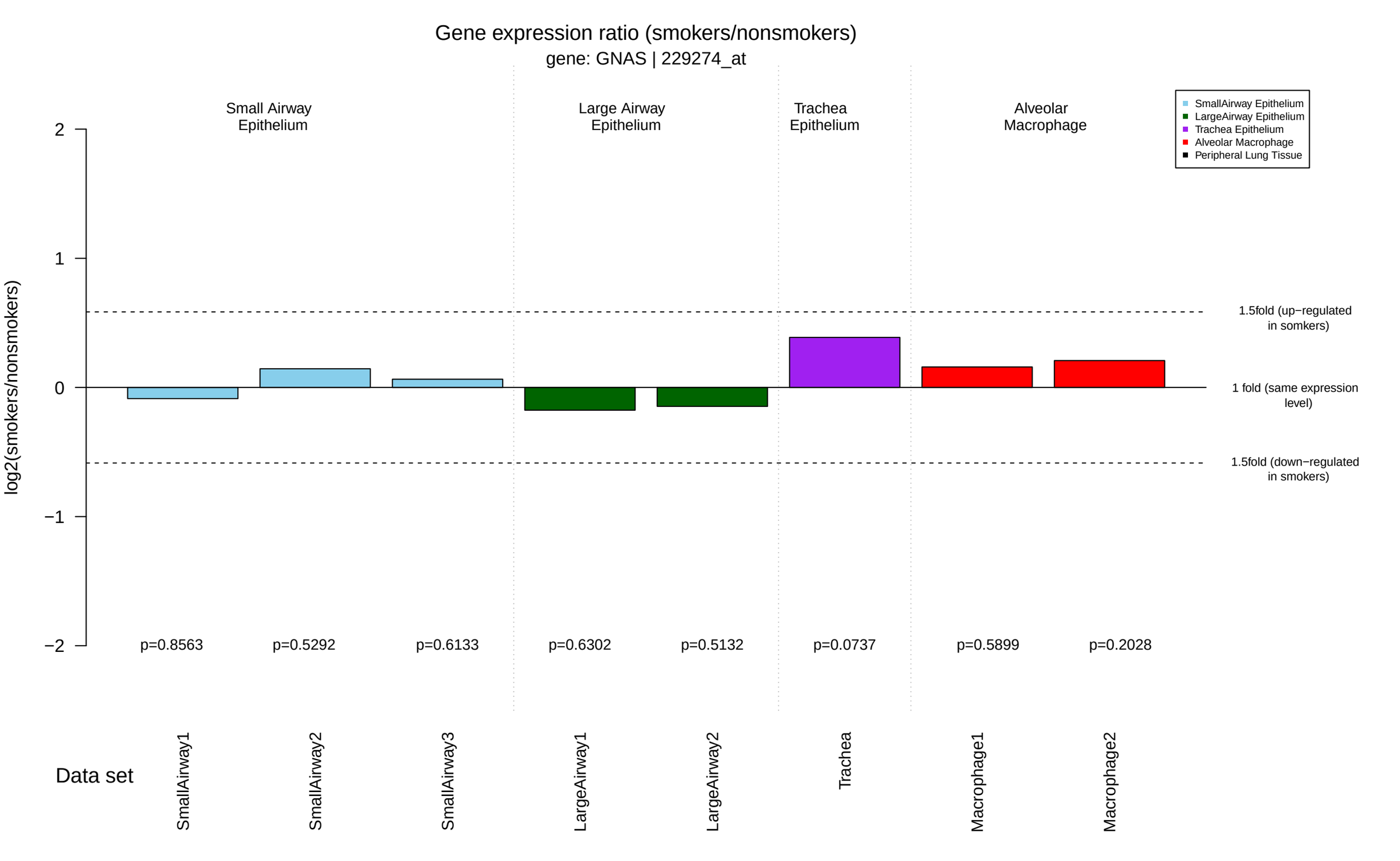
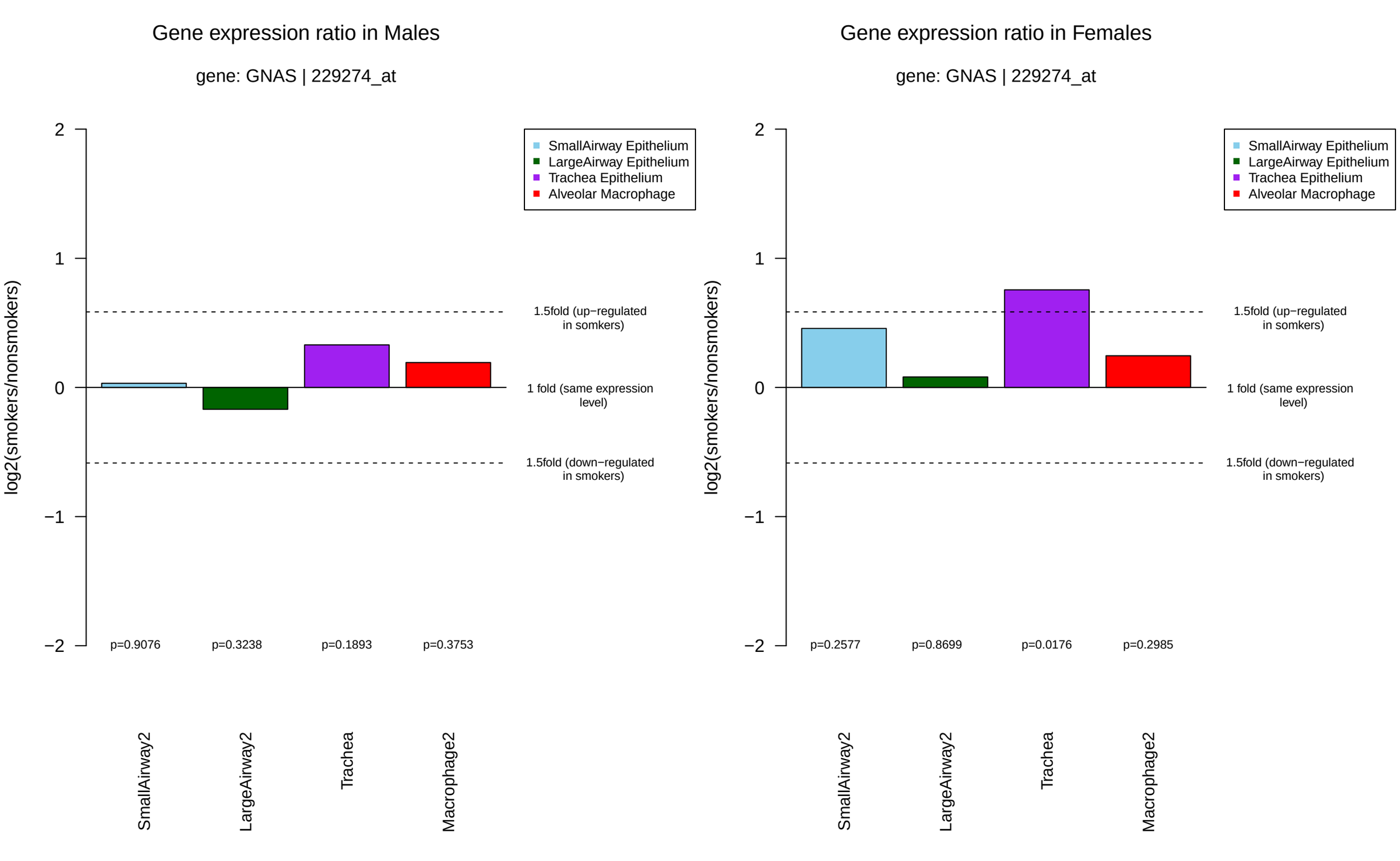
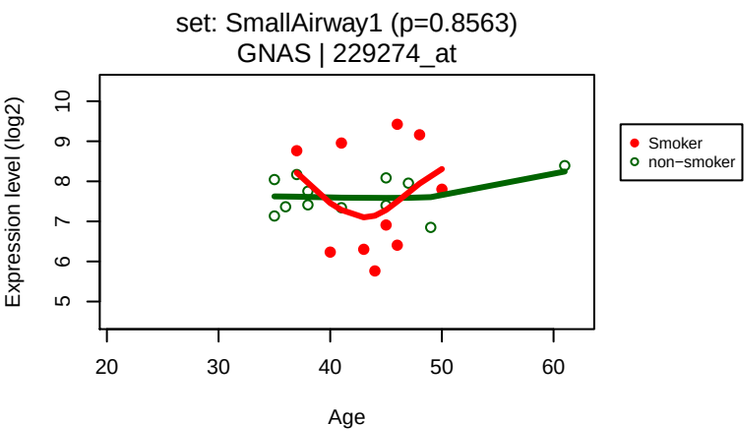
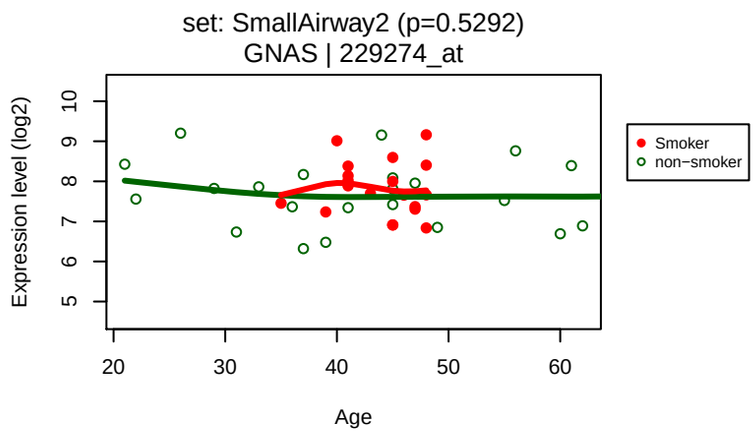
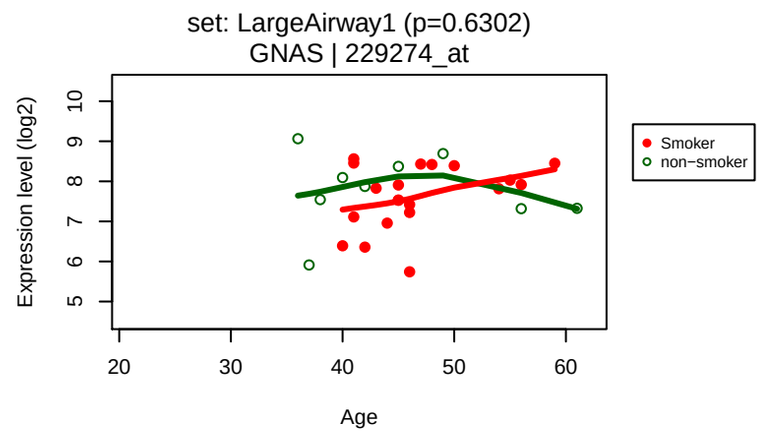

Gene expression of smokers and nonsmokers
gene: GNAS | 229274_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8563
p=0.5292
p=0.6133
p=0.6302
p=0.5132
p=0.0737
p=0.5899
p=0.2028
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

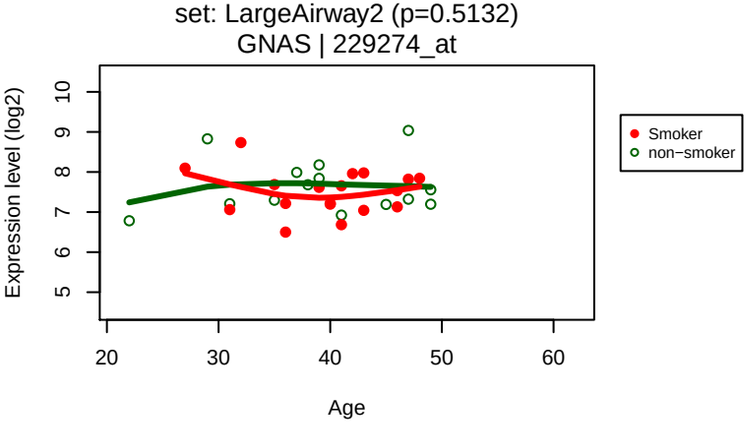

Gene expression ratio (smokers/nonsmokers)
gene: GNAS | 229274_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8563
p=0.5292
p=0.6133
p=0.6302
p=0.5132
p=0.0737
p=0.5899
p=0.2028
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

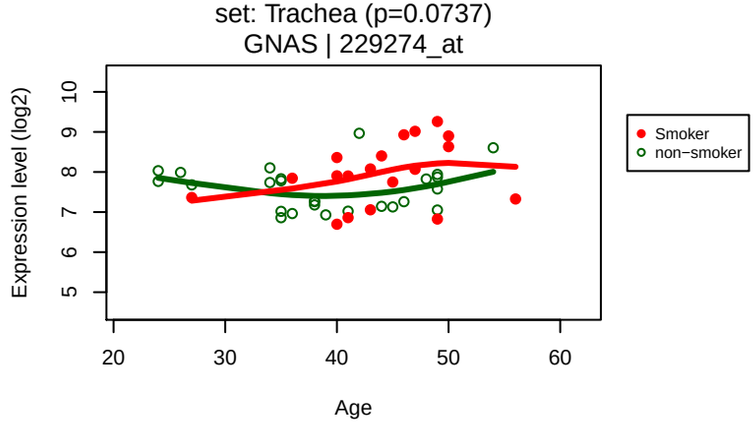

Gene expression ratio in Males
gene: GNAS | 229274_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9076
p=0.3238
p=0.1893
p=0.3753
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNAS | 229274_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2577
p=0.8699
p=0.0176
p=0.2985
SmallAirway2
LargeAirway2
Trachea
Macrophage2

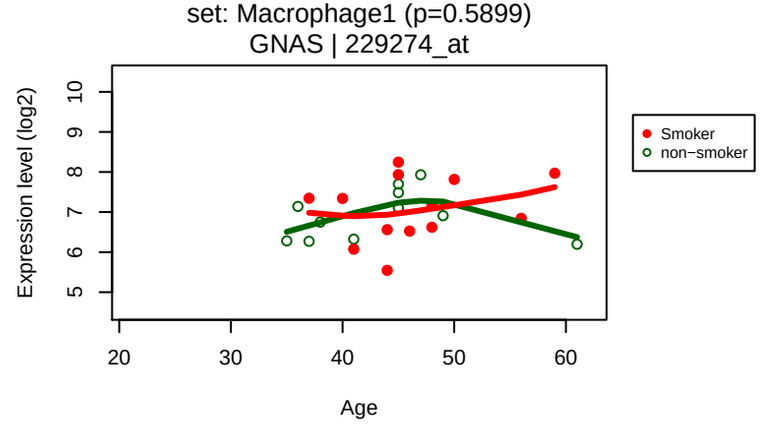

set: SmallAirway1 (p=0.8563)
GNAS | 229274_at
Expression level (log2)
Age
Smoker
non−smoker

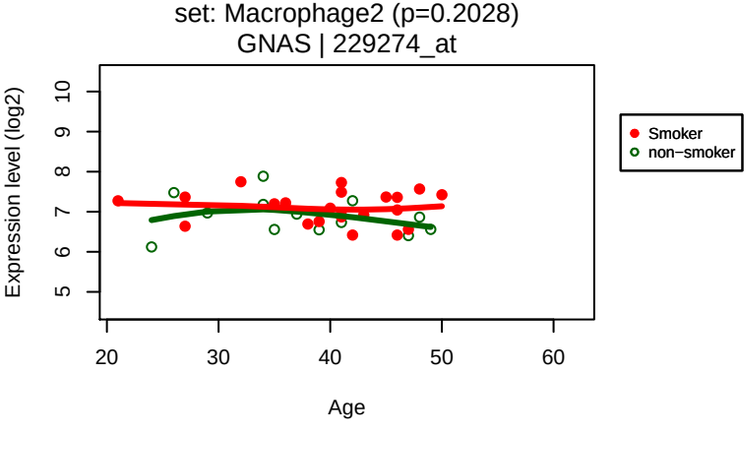

set: SmallAirway2 (p=0.5292)
GNAS | 229274_at
Expression level (log2)
Age
Smoker
non−smoker

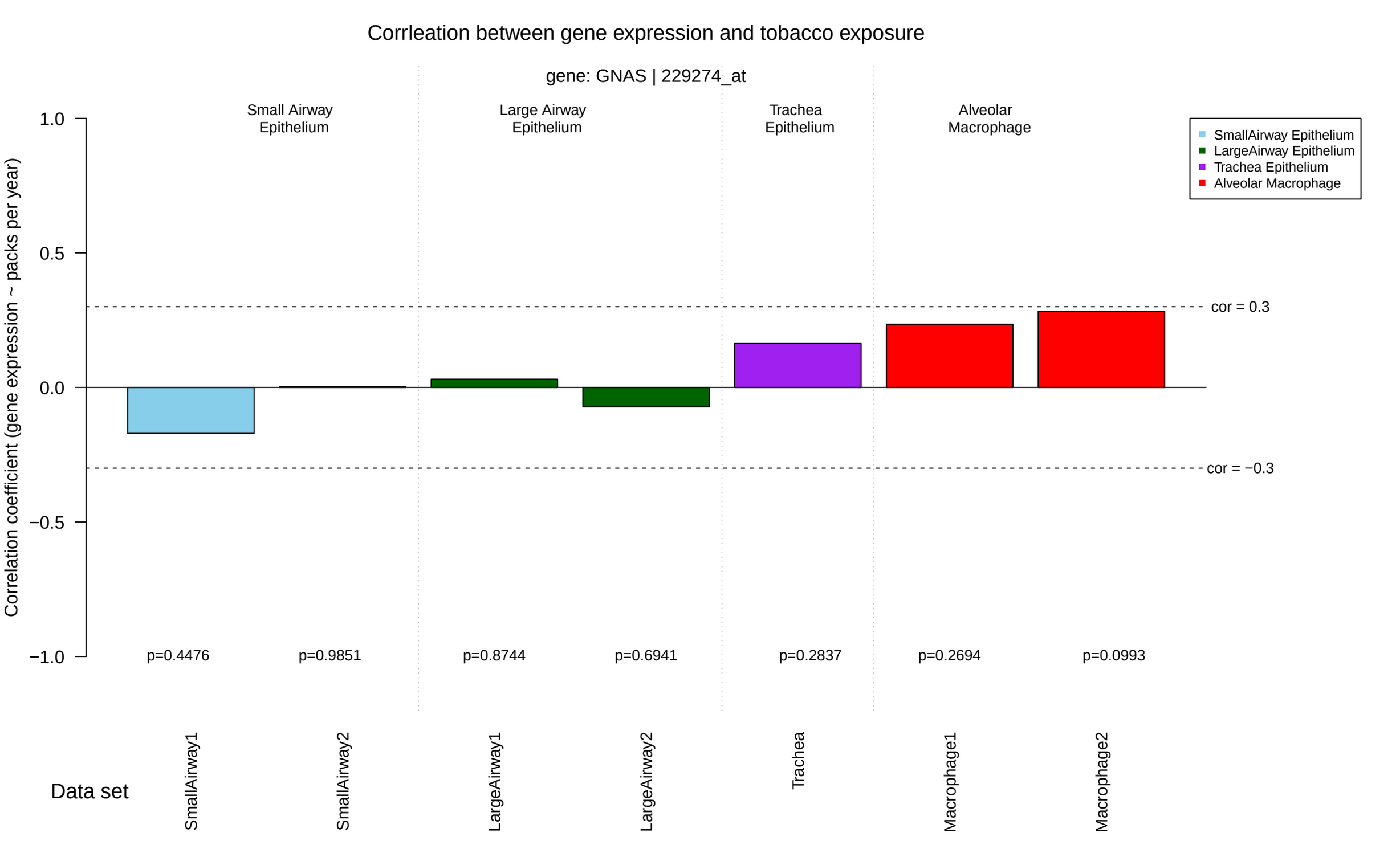

set: LargeAirway1 (p=0.6302)
GNAS | 229274_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5132)
GNAS | 229274_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0737)
GNAS | 229274_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5899)
GNAS | 229274_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2028)
GNAS | 229274_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNAS | 229274_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.4476
p=0.9851
p=0.8744
p=0.6941
p=0.2837
p=0.2694
p=0.0993
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2